

Module - 2Materials for Memory and Display system.

Memory Device: A memory device is a piece of hardware used to store data. Most electronic devices such as computers, mobile phones, tablets, etc all have a storage device that stores data and I/O programs.

Basic concepts of Electronic memory:

An electronic memory device is a form of semiconductor storage which is fast in response and compact in size.

A Semiconductor storage system which can be read and written when coupled with a central processing unit (CPU, a processor).

The basic goal of a memory device is to provide a means for storing and accessing binary digital data sequences of '1's and '0's.

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Electronic memory devices consists of

- 1) Two electrodes
- 2) Switching layer between two electrodes.

The layer is operated from High Resistance state (HRS) to Low resistance state (LRS) under an external electric voltage.

The HRS can be regarded as "0" bit in data storage (OFF)

The switching from HRS to LRS is equivalent to "0" to "1" binary conversion (ON).

If a single material (used in making memory device) provides more than two resistance states (bistable), the storage capacity of a single memory increases exponentially.

Classification of Electronic Memory Devices

Electronic memory devices are classified into 4 types

- 1) Transistor - Type Electronic memory
- 2) Capacitor - Type Electronic memory

3) Resistor - Type Electronic Memory

4) Charge Transfer effects.

1) Transistor - Type Electronic Memory

Transistors are made from Silicon (it is a semiconductor). It is converted to p-type and n-type semiconductor by doping trivalent and pentavalent impurities. Transistors are made using p-type and n-type semiconductor. A transistor is a miniature electronic component that can work either as an amplifier or as a switch.

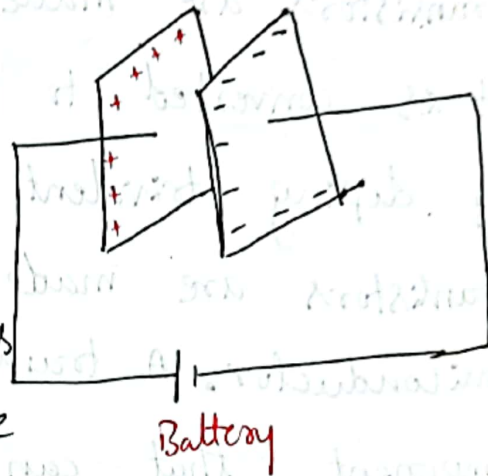
A computer memory chip consists of billions of transistors, each transistor is working as a switch, which can be switched ON or OFF. Each transistor can be two different states and store two different numbers, Zero and ONE. Since chip is made of billions of such transistors and can store billions of zeros and ones, and almost every number and letter can be stored.

2) capacitor type electronic memory:

A capacitor consists of two metal plates which are capable of storing an electric charge. It is used to store data. It is like a battery that holds data based on energy.

If the parallel plates of a capacitor are

It the capacitor is charged, it holds the binary numerals "1" and holds "0" when the cell is discharged.



If the parallel plates of a capacitor are separated by dielectric layer, charges dissipate slowly and memory would be volatile.

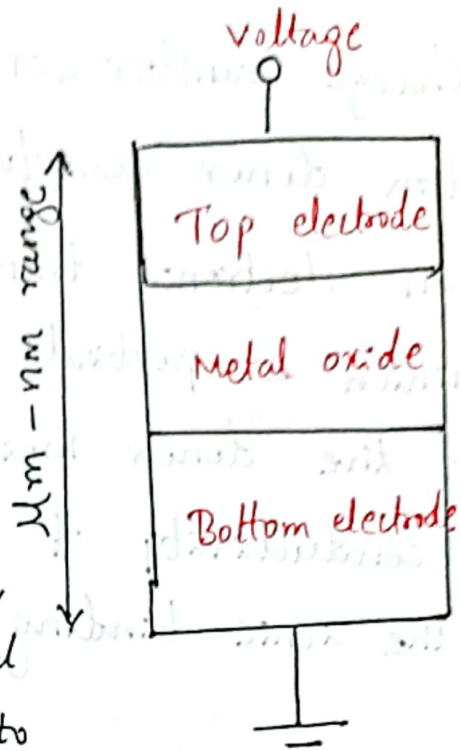
on the other hand, if the medium between the electrodes is ferroelectric in nature, can maintain permanent electric polarisation that can be repeatedly switched between two stable states (bistable) by an external electric field.

Thus memory based on ferroelectric capacitors (FeRAM) is non-volatile memory.

3) Resistor - Type Electronic memory:

Memory devices containing switchable resistive materials are classified as resistor-type memory @ resistive random access memory (RRAM).

Resistor-Type electronic memory usually has a simple structure, having a metal-insulator-metal structure generally referred to as MIM structure.



The structure comprises of an insulating layer (I) sandwiched between the two metal (M) electrodes and supported on a substrate (glass, silicon wafer, plastic @ metal foil).

Initially, the device is under high resistance state @ OFF and logically "0" state, when resistance changed @ under external applied field changes to low resistance state @ "ON" logical value "1".

4) Charge Transfer effects:

A charge transfer (CT) complex is defined as an electron donor-acceptor (D-A) complex, characterised by an electronic transition to an excited state in which a partial transfer of charge occurs from the donor moiety to the acceptor moiety. The conductivity of a CT complex is dependent on the ionic binding between the D-A components.

Classification of electronic memory based on storage type of the device

Electronic memory based on storage device mainly classified into volatile and non volatile memory.

Classification of organic memory Devices

organic memory molecules materials are classified as.

- 1) organic molecules
- 2) polymeric materials
- 3) organic-Inorganic hybrid materials.

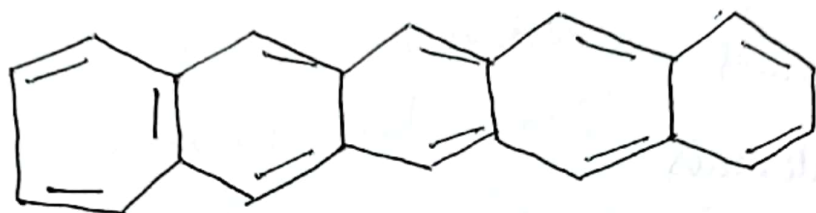
1) organic molecules:

There are different category of organic molecules which show bistable @ multistable states when external field is applied. when a threshold voltage is applied they undergo a transition from OFF state to the ON state, @ ON state to the OFF state. All these materials can be used in organic electronic memory devices.

Acene derivatives are the first compounds to find applications in memory devices.

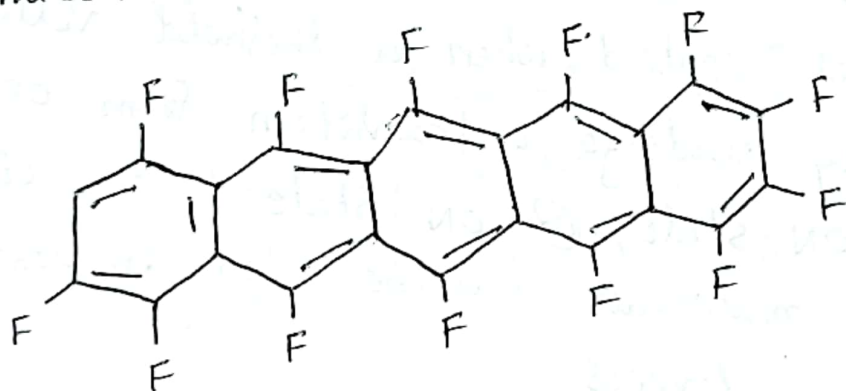
Acene derivatives are the polycyclic aromatic compounds consisting of linearly fused benzene rings. The best example is pentacene, in which five benzene rings are fused linearly.

It can be synthesized in thin film @ crystal form. Both forms exhibits a very good hole mobility and hence it behaves as a p-type semiconductor.



Pentacene

The perfluoropentacene, obtained by substituting all 'H' atoms by 'F' atoms, behaves as n-type semiconductor due to high electronegativity of F atoms.



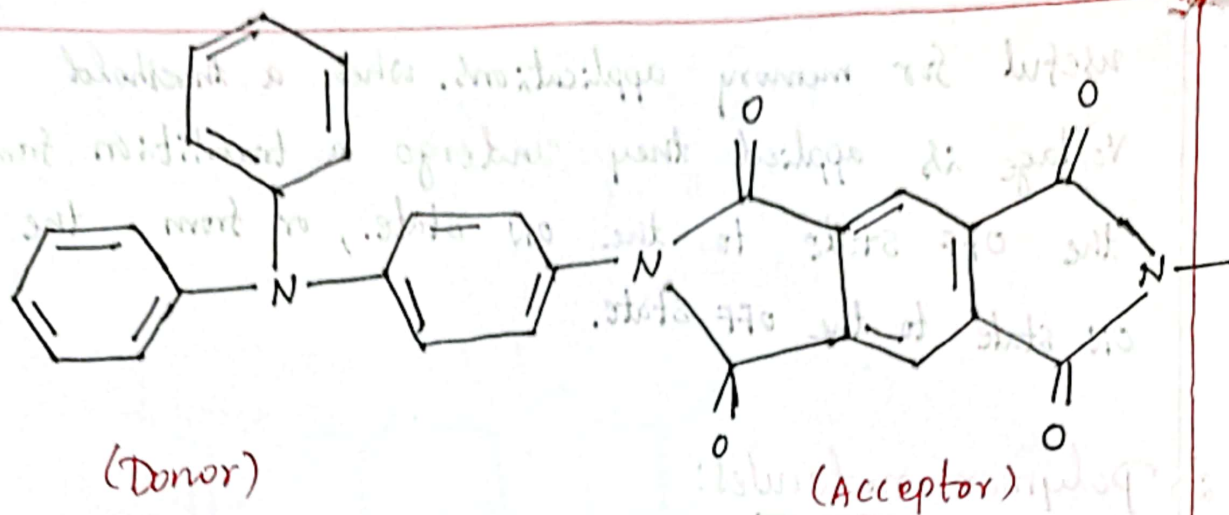
perfluoropentacene

Thus, pentacene and perfluoropentacene, both have similar crystal packing but former behaves as p-type semiconductor and latter behaves as an n-type semiconductor. Therefore, these molecules together exhibit charge transfer processes that are

useful for memory applications. When a threshold voltage is applied they undergo a transition from the OFF state to the ON state, or from the ON state to the OFF state.

2) Polymer molecules:

Functional polyimides (PIs) are one of the most commonly used polymeric materials for organic electrical memory applications. They have high thermal stability ($>500^{\circ}\text{C}$) and mechanical strength and can be easily processed from solution. In functional PIs, phthalimide acts as the electron acceptor, and triphenylamine acts as an electron donor to form a Donor-Acceptor structure. They exhibit two stable charge states under applied electric field. These states arise due to transfer of electrons from donor to acceptor. This bistability is used to store data in memory device.



3) organic - Inorganic Hybrid materials

Generally, organic - Inorganic hybrid materials are composed of organic layers containing inorganic materials. Inorganic materials used are allotropes of carbon like fullerenes, carbon nanotubes, graphene and metal nanoparticles, semiconductor nanoparticles and inorganic quantum dots (QDs).

(1) organic - carbon Allotrope Hybrid Materials

Polymers containing electron donors, such as thiophene, fluorene, carbazole and aniline derivatives can be combined with fullerenes to obtain a charge transfer hybrid material with donor - Acceptor ability and electrical bistable states. Fullerenes exhibit high electron-withdrawing ability, and

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can capture up to six electrons. They are used in WROM memory effect devices.

(2) Organic - Inorganic Nanocomposites

These are hybrid electronic memory devices in which organic polymer with appropriate functional group is clubbed with metal nanoparticles. An example is a composite of 8-hydroxyquinoline containing polymer with gold nanoparticle sandwiched between two metal electrodes. Bistable electronic transition states are observed when an electric field is applied due to charge transfer between the Au nanoparticles and 8-hydroxyquinoline.

The Au nanoparticles and 8-hydroxyquinoline containing polymer with gold nanoparticle sandwiched between two metal electrodes. Bistable electronic transition states are observed when an electric field is applied due to charge transfer between the Au nanoparticles and 8-hydroxyquinoline.

History of ^{organic} polymer Electronic Memory Devices

Polymer electronic memory devices are a type of memory technology that uses polymer materials as the active layer for storage.

Polymer electronic memory devices are still in the early stages of development, but they have the potential to offer a new type of memory technology with unique properties and capabilities. Some of the potential applications of polymer memory devices include flexible electronics, wearable devices.

* In 1970s, some polymers such as PMMA, PEMA and PS were reported to show switching effects.

* In 1980s, thin films of polymer materials capable of exhibiting non-volatile memory effects were invented.

* In 1990s, development of polymer ferro-electric random access memory (FeRAM).

* In 2000s, an organic field-effect transistors (OFETs) memory device was developed.

* Development of numerous charge transfer (CT) complexes with an electron donor and an electron acceptor.

exhibiting bistable electrical switching and memory effect.